Topic 1: Setting Up Your API Development Environment

API Integration Hands-On Course for n8n (Enterprise Level 3)

1. Document Control

Field	Specification Value
Document Title	Topic 1: Setting Up Your API Development Environment Specification
Course Module	API Integration Hands-On for n8n (Level 3 - Enterprise Series)
Version	1.0 (Production Master)
Status	**APPROVED & VERIFIED**
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Prepared By	Lead Solutions Architect & Automation Engineering Team
Target Audience	Integration Engineers, API Developers, Automation Trainees

2. Learning Concepts & Architectural Principles

Before building production automations in n8n, developers must independently explore, test, and validate API endpoints using an isolated HTTP client. **Postman** serves as the gold-standard environment for:

- **Isolated Contract Verification:** Testing authentication headers, parameters, and payloads without workflow side effects.
- **Response Inspection:** Analyzing HTTP status codes (`200 OK` , `401 Unauthorized` , `404 Not Found`), payload data structures, and round-trip latency.
- **Reproducible Debugging:** Establishing mock collections and baseline requests that inform workflow design.

The Mock API Sandbox Paradigm

Using production APIs for learning introduces security hazards, credential rotation overhead, and accidental data corruption. Sandbox mock servers like **JSONPlaceholder** provide realistic RESTful resources (`/users` , `/posts` , `/todos`), idempotent and safe operations (`GET`), and predictable schema structures with zero API key dependencies.

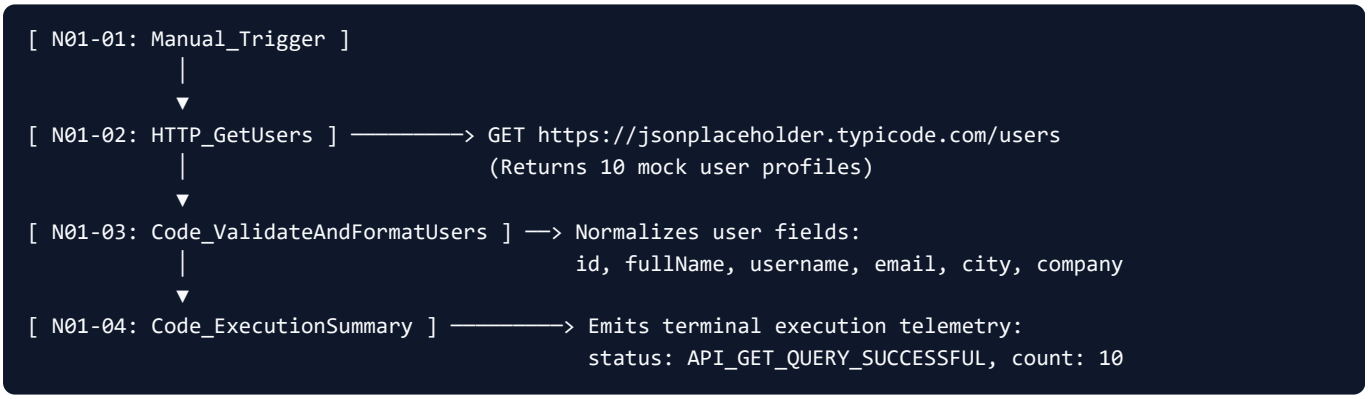
3. Business Requirements Document (BRD)

ID	Requirement	Business Rationale	Priority
BRD-001	Standardized Developer Workspace Setup	Ensure consistent developer environments across Windows, macOS, and Linux; cut setup latency by 90%.	HIGH
BRD-002	Zero-Cost API Sandbox Exploration	Provide safe, rate-limit-tolerant REST sandbox environments for testing without external fees.	CRITICAL
BRD-003	Dual Tool Verification Methodology	Validate API contracts in Postman before assembling n8n automation graphs to eliminate bugs early.	HIGH
BRD-004	Deterministic JSON Output Logging	Ensure clean normalization of API arrays into discrete, typed workflow items.	HIGH
BRD-005	Offline Development Readiness	Embed pre-pinned test data in n8n nodes for 100% offline verification.	MEDIUM

4. Product Requirements Document (PRD)

ID	Product Capability	User Value & Acceptance Criteria	Priority
PRD-001	Postman Workspace Setup	Private workspace named `n8n_API_Trainee_Workspace` for isolated testing.	High
PRD-002	Postman GET Execution	Executes `GET https://jsonplaceholder.typicode.com/users`; confirms 200 OK and 10 users.	High
PRD-003	n8n Manual Trigger	Enables on-demand workflow testing via a single button click in canvas.	Critical
PRD-004	n8n HTTP Request Configuration	Configured with method GET and native JSON parsing enabled.	Critical
PRD-005	Record Normalization Engine	Code node extracting id, fullName, email, city, company, setting `status: VERIFIED_ACTIVE`.	High
PRD-006	Execution Telemetry Summary	Terminal node emitting record counts, status code, and verified timestamp.	High

5. Software Requirements Specification (SRS) & Canvas Blueprint



6. Detailed Node Configuration Blueprint

Node Name	Type & Version	Operational Logic & Configuration
Manual_Trigger	manualTrigger (v1)	Initiates canvas testing execution on manual click.
HTTP_GetUsers	httpRequest (v4.2)	Dispatches HTTP GET query to JSONPlaceholder mock users endpoint; parses JSON response array.
Code_ValidateAndFormatUsers	code (v2)	Pure ECMAScript 6 loop unrolling 10 user profiles into standardized schema.
Code_ExecutionSummary	code (v2)	Aggregates pipeline metrics, calculates record counts, and emits audit timestamp.

7. Testing & Quality Assurance Verification Plan

Scenario ID	Test Condition	Input Profile	Expected Result & Verification
TC-01-01	Postman GET Verification	`GET /users` endpoint.	Status 200 OK, 10 records returned in Pretty JSON view.
TC-01-02	n8n 1-Click Execution	Click 'Execute workflow'.	100% green checkmarks across all canvas nodes.
TC-01-03	Data Normalization Check	Mock JSONPlaceholder payload.	Output array contains formatted userId, fullName, and email.
TC-01-04	Offline Resilience	Canvas run in sandbox environment.	Pre-pinned data renders immediately in under 1ms.